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Project Title:	Fundamental Studies of Crystal Growth of Microporous Materials		
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Task Description:	Conventional synthesis of microporous materials usually involves mixing together reactants in an aqueous medium and results in immediate formation of an amorphous material, usually referred to as the gel from which crystal growth occurs. The complexity of the reaction mixture makes it difficult to examine the process. In our synthesis procedure, reactants are dispersed in small droplets of water in an oil-like medium with the help of detergents. These nanoclusters of water are called reverse micelles. The reactants in two different micelles can exchange with one another upon collision and if the conditions are appropriate, nucleation can occur. Our focus has been on the synthesis of zincophosphates, referred to as ZnPO. The first microporous material, zincophosphate sodalite was made using AOT reverse micelles. Considerable efforts were made to make other framework structures by primarily changing the Zn and phosphate concentrations in the AOT reverse micelles, and were unsuccessful. Recent experiments with the cationic reverse micelles DODMAC have led to the formation of open framework ZnPO-X, the structural analogue of the important gasoline cracking catalysts (zeolite Y or topoplogy). We have now broadened the scope of this synthetic method by synthesizing ZnPO-X with TOMAC reverse micelles. Our primary interest is a fundamental understanding of the molecular assembly of microporous materials. Can that goal be attained by examining crystal dissolution? We believe that to be the case for several reasons. First, the most interesting microporous frameworks are metastable structures and evolve to more condensed topologies with time. Second, clues regarding the complexity of the dissolution fragments of different frameworks provide information about the stability of the building units. There are two aspects of microgravity that we want to exploit. These include lack of sedimentation and absence of buoyancy driven convection. We have demonstrated that, under particular conditions of reverse micelle composition, crystal growth occurs in a layer-by-layer fashion. Avoiding sedimentation in these systems will lead to the growth of larger crystals. Light scattering will be used to follow the growth of crystals in microgravity. Seeding experiments to grow larger crystals is also possible, something that has not been accomplished in conventional synthesis. The second aspect of the microgravity experiments will focus on microprobe Raman studies, and absorption/fluorescence microscopy based on pH sensitive dyes. By comparing results in 1g and microgravity, we will discover if different surface intermediates are present during dissolution of different frameworks. Microporous materials include a large group of solids of varying chemical composition as well as porosity. These materials are characterized by channels and cavities of molecular dimensions. The framework structure is made up of interconnecting T-O-T' bonds, where T and T' can be Si, Al, P, Ga, Fe, Co, Zn, B and a host of other elements. Materials with Si-O-Al bonding in the framework are called zeolites and are extensively used in many applications. Ion-exchange properties of these materials are exploited in the consumer and environmental industries. Chemical and petroleum industries use zeolites as catalysts in hydrocarbon transformations. Synthesis of new microporous frameworks has led to the development of new technologies, and thus considerable effort worldwide is expended in their discovery. The complexity of the synthesis process and limited understanding of it has made it difficult to develop directed synthesis of microporous materials and most advances in this field have been made by trial and error. We are addressing the following basic issues in crystal growth of these materials including - Nature of the nucleation process - Molecular structure and assembly of nuclei - Growth of nuclei into crystals - Morphology control - Transformation of frameworks into other structures. Microporous materials are crystalline frameworks with well-defined cages and channels of molecular dimensions varying from 2 to 16 Angstroms. Various elements such as silicon, aluminum, zinc, phosphorus, gallium and transition metals can form the backbone of the framework. The most commonly used microporous frameworks are aluminosilicate zeolites. These find extensive use in detergents and catalysts. The ion-exchange of the extra framework ions is exploited in detergent use. The catalytic behavior arises because superacidity can be introduced into these frameworks and the topography of the channels and cages influence product formation. Discovery of novel frameworks can have significant impact on practical applications, as in the recent introduction by Unilever of a new zeolite in detergents and the use of ZSM-5 in petroleum cracking. Thus, there is considerable effort worldwide in the synthesis of new frameworks. Our contributions in understanding the basic molecular events that are responsible for formation and dissolution of microporous materials should eventually lead to more directed synthesis of novel framework structures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study has provided basic information on the mechanism of growth of microporous materials and application of nanometer sized microporous crystals.
	A) Crystal growth from reverse micelles: The surfaces of zeolite crystals grown using reverse micelles showed significant differences then those microporous crystals synthesized via conventional hydrothermal method. Besides, this advantage, the reverse micelle method provides a convenient system to monitor the effects of each synthesis parameter on the growth of zeolite nanocrystals. For understanding the role of high temperature on reverse micelle, the growth of gold nanoparticles which is a well studied system was taken as an example. The temperature was increased by microwave heating, which basically increases the local temperature in the water cores of the reverse micelles. This alteration of the growth conditions did in fact increase the size of the gold nanoparticles and appeared to also promote a higher degree of crystallinity. Both the size and crystallinity of the gold nanoparticles were determined using transmission electron microscopy (TEM). Based on this observation, further studies are in progress to alter the growth conditions for the synthesis of larger and more crystalline microporous zincophosphate using microwave irradiation. B) The directing Role of 1,4-diazadicyclo[2,2,2] octane (DABCO)-Phosphate unit in the synthesis of zincophosphate Faujasite (ZnPO-X) The sequence of addition of various reactants during the synthesis of zeolites, plays an important role during the nucleation and growth stages. During the synthesis of microporous zincophosphate, the nucleation occurred very rapidly, within a minute of mixing of the reactants. The hypothesis was that the presence of DABCO – phosphate units in the synthesis medium accelerates the nucleation of ZnPO. The experimental strategy involved analysis of a combinatorial mixture of the reactants DABCO, phosphoric acid, zinc nitrate and sodium hydroxide. The system that had the DABCO and phosphate in the same solution nucleated faster then when DABCO and phosphate were added from different solutions. This was because DABCO and phosphate unit when together were found to form a

Task Progress:	"pre-structure" which acted similar to a template that allows for quick nucleation and growth of zincophospate X (ZnPO-X). Further, a DABCO-phosphate salt was also crystallized from reaction mixtures typical for the formation of ZnPO-X (in the absence of Zn2+) and its structure was determined by single crystal XRD. The rate of synthesis of ZnPO-X was faster by mixing this solid DABCO-phosphate salt than when phosphoric acid and DABCO were added separately. The results were further substantiated by molecular modeling studies. A paper has been published on this work in Microporous and Mesoporous Materials Vol.71, 1-3, 149-155, 2004. C) Memory effect of partially formed nuclei in the zeolite growth : Synthesis of nano zeolites have been carried out from clear solution by many researchers in order to understand the early events that lead to nucleation and subsequent zeolite growth. However, it has been found that the yields of zeolites from such compositions are quite low, indicating that significant amount of reactants are left behind in solution. Further, even midsynthesis addition of reactants could lead to secondary nucleation indicating that primary building units may be present. These two observations imply that the mother liquor recovered after the crystallization should contain the primary building units or structures that can readily evolve into these building units. In order to manifest these observations we examined the mother liquor left behind after the completion of the crystallization of the nanocrystalline sodalite, zeolite A and Y from clear solutions. The mother liquor solutions were lypholized and examined by STEM. For few regions in the mother liquor of LTA and FAU, the electron diffraction pattern clearly suggested crystalline regions of about 2-5nm. Further, many researchers speculated that the primary building units are distinct for each zeolite framework. We reasoned that if the primary building units are different, then solutions containing these units should exhibit a memory effect. We investigated if these mother liquors recovered from the nanocrystalline zeolite synthesis had any memory of the zeolites and can it direct the same compositions to different frameworks? Our synthesis experiments involved addition of small amounts of mother liquor from which crystal growth has occurred to different compositions, with an intention not to alter the ratio of various components in these compositions, but enabling it to direct these compositions to the specific zeolite. Composition A 15.19(TMA)2O: 0.63 Na2O:1.6Al2O3: 44.9SiO2: 805H2O could be directed to sodalite or zeolite A, when the corresponding mother liquor was added. Thus, demonstrating that the solutions recovered from synthesis of nanocrystalline zeolites manifest a memory effect. However, with FAU mother liquor, composition A did not lead to zeolite Y, whereas compositions B (13.68 (TMA)2O:~0.05Na2O:2.188Al2O3:~9.34SiO2:805H2O) and C(~6.48 (TMA)2O:~0.05Na2O:2.188Al2O3:~9.34SiO2:805H2O) did. This suggests that in order for the species in mother liquor to manifest a memory effect, the nutrient composition is important. An interpretation of this observation is that the nutrient solutions need to provide species that act as the glue to combine the primitive units present in the mother liquor to viable nuclei. This hypothesis is also supported by dynamic light scattering studies. The growth studies of SOD mother liquor when added to Sodalite composition exhibit two growth processes. One occurred at ~20 hours and the second at~40 hours. The second part of the growth curve is what is expected for sodalite growth. The first part of the growth curve is arising from the primitive units present in the mother liquor and takes about 20 hours for crystal growth immaterial of composition to which it has been added. This indicates that the units in the mother liquor that grow into crystals are partially preformed and takes ~20 hours to grow into viable nuclei. Thus, this study shows that the particles that are left behind in the nanocrystalline zeolite synthesis do possess a memory and can grow to the corresponding nuclei if provided with the right building units.
Bibliography Type:	Description: (Last Updated: 03/10/2008)
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Articles in Peer-reviewed Journals	Ramsharan Singh, John Doolittle, Jr. and Prabir K. Dutta, "The Directing Role of 1,4-diazabicyclo[2,2,2]octane (DABCO)-Phosphate Unit in Synthesis of Zincophosphate Faujasite (ZnPO-X), " Micro. & Meso. Mater, 71(1-3), 149-155, 2004. , Jan-2004